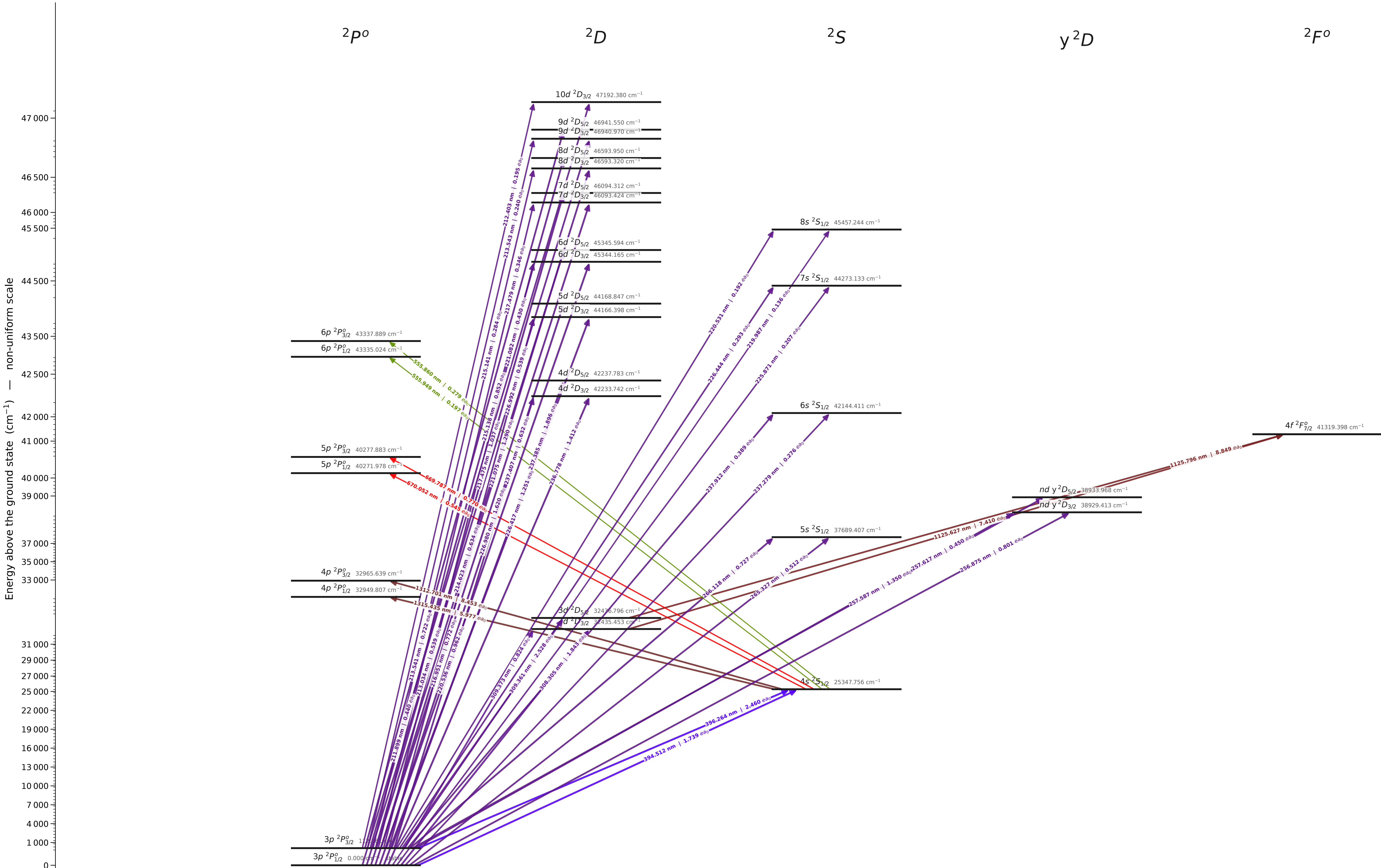
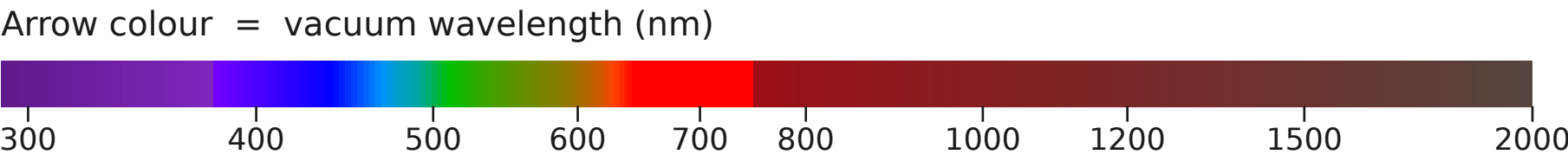


Aluminium

Al (Al I). Strongest classified transitions below 2 μm (NIST): 32 levels, 44 lines.

Al⁺ ionisation limit 48278.48 cm⁻¹ = 5.98577 eV (207.132 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle U||er||J'\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U er J'\rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$3p\ ^2P^o_{1/2} - 8d\ ^2D_{3/2}$	214.6230	214.5555	1396.8326	0.634	2.060e+07	-	-	-
$3p\ ^2P^o_{3/2} - 8d\ ^2D_{5/2}$	215.1376	215.0699	1393.4920	0.852	2.460e+07	-	-	-
$3p\ ^2P^o_{1/2} - 7d\ ^2D_{3/2}$	216.9507	216.8827	1381.8461	0.772	2.960e+07	-	-	-
$3p\ ^2P^o_{3/2} - 7d\ ^2D_{5/2}$	217.4752	217.4071	1378.5132	1.037	3.530e+07	-	-	-
$3p\ ^2P^o_{1/2} - 6d\ ^2D_{3/2}$	220.5355	220.4668	1359.3839	0.962	4.370e+07	-	-	-
$3p\ ^2P^o_{3/2} - 6d\ ^2D_{5/2}$	221.0749	221.0060	1356.0672	1.290	5.200e+07	-	-	-
$3p\ ^2P^o_{1/2} - 5d\ ^2D_{3/2}$	226.4165	226.3464	1324.0753	1.251	6.830e+07	-	-	-
$3p\ ^2P^o_{3/2} - 5d\ ^2D_{5/2}$	226.9798	226.9096	1320.7892	1.620	7.580e+07	-	-	-
$3p\ ^2P^o_{1/2} - 4d\ ^2D_{3/2}$	236.7775	236.7052	1266.1357	1.412	7.610e+07	-	-	-
$3p\ ^2P^o_{3/2} - 4d\ ^2D_{5/2}$	237.3847	237.3122	1262.8974	1.896	9.070e+07	-	-	-
$3p\ ^2P^o_{3/2} - nd\ y^2D_{5/2}$	257.5865	257.5094	1163.8515	1.350	3.600e+07	-	-	-
$3p\ ^2P^o_{3/2} - 5s\ ^2S_{1/2}$	266.1178	266.0386	1126.5405	0.727	2.840e+07	-	-	-
$3p\ ^2P^o_{1/2} - 3d\ ^2D_{3/2}$	308.3046	308.2151	972.3904	1.843	5.870e+07	-	-	-
$3p\ ^2P^o_{3/2} - 3d\ ^2D_{5/2}$	309.3606	309.2708	969.0712	2.528	7.290e+07	-	-	-
$3p\ ^2P^o_{1/2} - 4s\ ^2S_{1/2}$	394.5122	394.4006	759.9066	1.739	4.990e+07	-	-	-
$3p\ ^2P^o_{3/2} - 4s\ ^2S_{1/2}$	396.2641	396.1520	756.5471	2.460	9.850e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).